CVPR 2025 · ORAL

AIpparel

A Multimodal Foundation Model for Digital Garments

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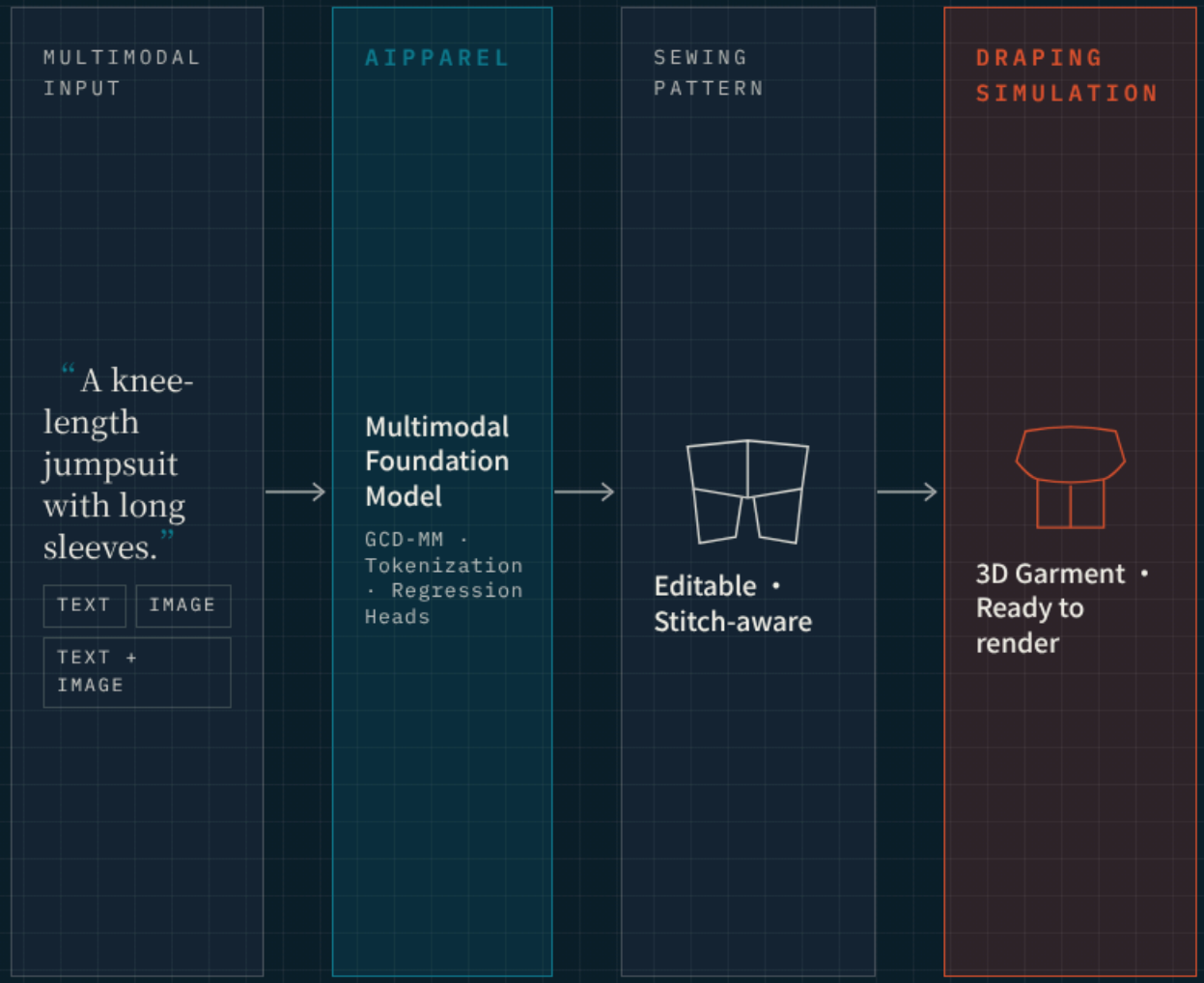
Stanford University • ETH Zürich

IEEE/CVF CVPR 2025

georgenakayama.github.io/AIpparel

FIG. 1 · PIPELINE

Multimodal Input → Sewing Pattern → Draping Simulation



TEXT-TO-GARMENT · TEXT-AND-IMAGE-TO-GARMENT · IMAGE-TO-GARMENT · GARMENT EDITING

AGENDA

What I'll cover

Five beats, twelve minutes

01 **Background & motivation**
Why garment creation is still a manual bottleneck.

02 **Limitations of existing methods**
Single-modality focus, data scarcity, complexity ceiling.

03 **Method**
GCD-MM dataset, new tokenization, multimodal model.

04 **Experiments & results**
Image-to-garment, multimodal, editing — vs. SewFormer & DressCode.

05 **Takeaways & future work**
What we learned, and where the model could go next.

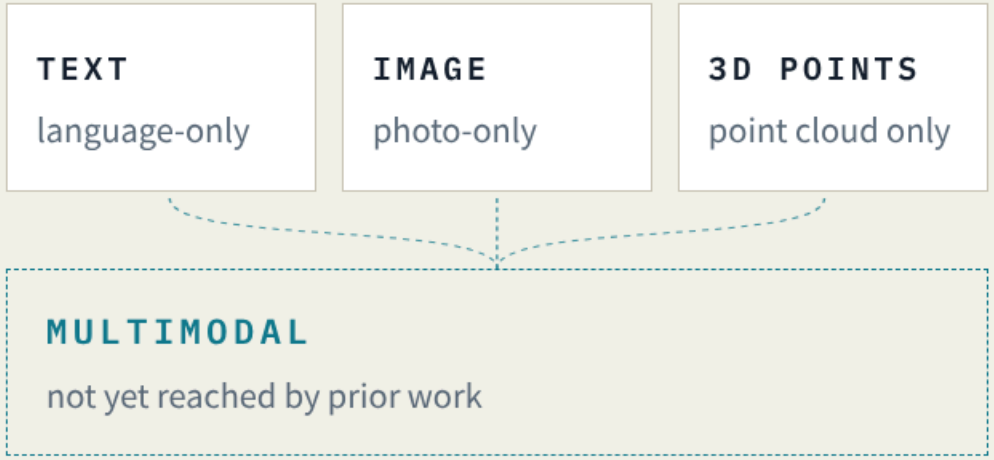
A roadmap in the order of the paper — each beat maps to one section of the CVPR submission.

§ CVPR 2025 · AIPPAREL

BACKGROUND

Pattern-making is a manual craft — and the models are single-modal

Two challenges block scalable multimodal garment generation



01 No large-scale multimodal sewing-pattern dataset is publicly available.

02 Accurate multimodal interpretation only emerges in models with billions of parameters.

WHY GARMENTS MATTER

Clothing serves as a barrier against the elements, a reflection of societal norms, and a means of personal expression.

WHY PATTERN-MAKING IS HARD

A sewing pattern is a set of flat 2D panels with standardized assembly instructions that form a complete 3D garment. The geometric relationship between the 2D panels and the draped 3D shape is complex and manual.

WHY EXISTING MODELS FALL SHORT

State-of-the-art sewing-pattern prediction methods are designed to work with one specific input modality — 3D points, images, or language — and do not compose across them.

LIMITATIONS

No prior dataset combines text, images, and edits at scale

Three limitations of the existing methods

TABLE 1 Modalities of Sewing Pattern Datasets

DATASET	TOTAL	TEXT	IMAGE	EDITS
WANG ET AL. [63]	8k	○	●	●
KOROSTELEVA & LEE [27]	23.5k	○	●	○
SEWFACTORY [36]	19.1k	○	●	○
DRESSCODE [21]	20.3k	●	○	○
GCD [30]	130k	○	●	○
GCD-MM OURS	120k	●	●	●

GCD-MM is a large-scale sewing pattern dataset with multimodal annotations, including text, images, and edited patterns.

01 **Single-modality focus**

Existing approaches are "often challenging to adapt to garment prediction tasks requiring different or combined input modalities."

02 **Data scarcity**

No prior dataset pairs text, images, and edited patterns at scale — every row above misses at least one modality.

03 **Complexity ceiling**

Methods such as Wang et al. are "limited to predicting simple garments with a predefined set of parameters."

FIG. 1 · COUNTER-EXAMPLE

"A knee-length jumpsuit with long sleeves."

AIpparel generates and simulates this complex silhouette from text alone — beyond the reach of prior single-modal methods.

METHOD

Fine-tune an LMM on a multimodal dataset with a new tokenization

Three contributions — dataset, tokenization, multimodal model

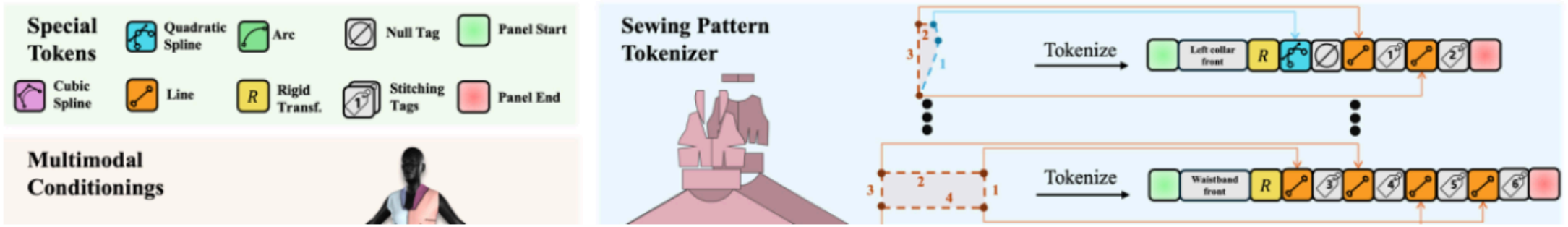


Figure 2 — Illustration of Our Method (from the paper, p4)

01 GCD-MM dataset

A multimodal sewing pattern dataset extending the largest public dataset of sewing patterns with multimodal annotations.

02 Novel tokenization

A new tokenization scheme and training objective for fine-tuning LMMs to predict sewing patterns.

03 Multimodal model

The first multimodal foundation model for sewing pattern prediction, taking language, images, and sewing patterns as input.

DATASET

GCD-MM — 120,000 garments with text, images, and edits

Built in three steps on top of GarmentCodeData

01

GCD base

Start from **GarmentCodeData** — a large collection of sewing patterns with rendered draped views.

02

Rule-based phrases + GPT-4o

A rule-based algorithm emits short phrases (e.g. “midi dress”, “flared hem”); GPT-4o is prompted with those phrases **plus** rendered views to write detailed natural-language descriptions.

03

Editing pairs

GarmentCode programming abstractions generate paired patterns with text instructions — longer skirt, add a hood, change neckline.

120,000
unique sewing patterns

10×
larger than prior SOTA datasets

90:5:5
train • validation • test split

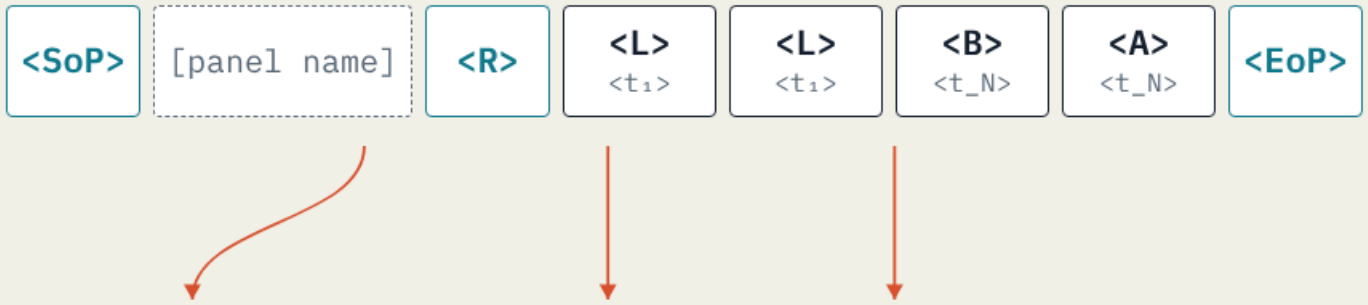
> “Our approach reduces GPT-4o’s hallucination and results in more accurate descriptions in natural language.”

TOKENIZATION

Special tokens + regression heads — 100× fewer tokens than DressCode

Eq. 1: E_g(P) = <SoG> E_p(P_1, S) … E_p(P_n, S) <EoG>

TOKEN STREAM — ONE PANEL two lines stitched together, one cubic Bézier curve, one arc



Positional embeddings project the ground-truth geometry back into the token representations: $h^{(e)}_{\phi} : \mathbb{R}^2 \to \mathbb{R}^D$ for the second edge endpoint, and $h^{(R)}_{\phi'} : \mathbb{R}^7 \to \mathbb{R}^D$ for the transformation parameters.

TOKENS PER GARMENT ON GCD-MM

250	838
average	maximum



~100× fewer tokens than DressCode to describe the same garment

TRAINING

Frozen vision encoder, fine-tuned LLM + regression heads

Mixed loss — cross-entropy for tokens, L2 for continuous parameters

EQ. 2 Fine-tuning loss

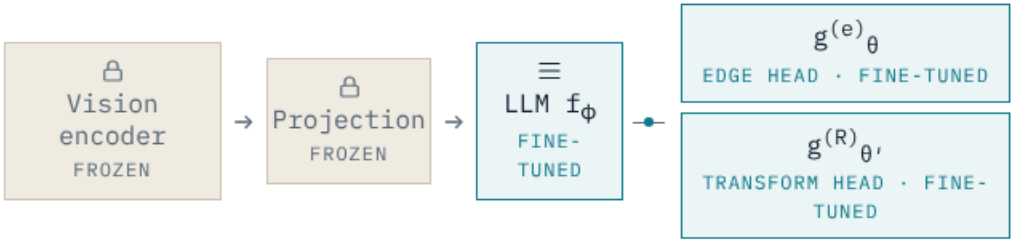
$$L = L_{CE}(f_\phi(X),\ y)$$
$$+ \lambda_e \sum_{e' \in E} \left\| g^{(e)}_\theta \circ f_\phi(X_{<i_{e'}}) - v^{e'}{}_2 \right\|^2$$
$$+ \lambda_R \sum_{R' \in R} \left\| g^{(R)}_{\theta'} \circ f_\phi(X_{<i_{R'}}) - R' \right\|^2$$

- **Cross-entropy**
  discrete token prediction
- **Edge L2**
  panel vertex endpoints v_2
- **Transform L2**
  rotation & translation R′

All L2 metrics are measured in centimeters, except rotation.

Architecture

LLaVA 1.5-7B · GCD-MM



Evaluation metrics

single model · GCD-MM test

1. **Panel L2 · Rot L2 · Transl L2**
   Average vertex, rotation and translation L2 distance between predicted and ground-truth panels.
2. **#Panel Accuracy**
   Percentage of sewing patterns with the correctly predicted number of panels.
3. **#Edge Accuracy**
   Percentage of correctly predicted edges within each correctly predicted panel.
4. **#Stitch Accuracy**
   Accuracy of predicted stitches compared to ground truth.

Reported Accuracy = #Panel Accuracy × #Edge Accuracy.

EXPERIMENTAL SETUP

Three tasks, two families of baselines

Image-to-garment • Multimodal generation • Language-driven editing

TASK	INPUT	BASELINES	METRIC FOCUS
01 **Image-to-Sewing-Pattern Prediction** §4.1	Single garment photo SewFactory · GCD-MM	• SewFormer • SewFormer-FT	**Panel L2** Accuracy = #Panel Acc × #Edge Acc
02 **Multimodal Garment Generation** §4.2	Text • Image • Text+Image • NL query • Edit instruction 100 garments · 5 input types · 20 samples each	• SewFormer-FT • **SewFormer-FT†** • DressCode • **DressCode†** augmented with GPT-4o + DALL-E	**Panel L2** #Panel / #Edge / #Stitch Accuracy
03 **Language-driven Sewing Pattern Editing** §4.3	Garment image + natural-language edit e.g. “add a hood”, “lengthen the skirt”	• DressCode • **DressCode*** • SewFormer • **SewFormer*** adapted with InstructPix2Pix + GPT-4o	**Panel L2** #Panel / #Edge / #Stitch Accuracy

● Base baseline ● Multimodal-augmented variant (†, *)

Multimodal benchmark: 100 garments × 5 input types × 20 samples

#Stitch Accuracy 77.2% vs 2.8% on GCD-MM

Table 2 — AIpparel vs SewFormer / SewFormer-FT

TABLE 2 · IMAGE-TO-GARMENT PREDICTION (QUANTITATIVE)

Lower is better for L2 · Higher is better for Acc

MODEL	PANEL L2	P-ACC	E-ACC	ROT L2	T L2	#STITCH ACC
SEWFACTORY — Simpler dataset — both models perform well						
SewFormer	3.3	89.8	99.3	.008	0.8	99.2
AIpparel	2.8	93.9	99.9	.005	0.6	99.8
GCD-MM — Complex garments — the decisive gap						
SewFormer-FT	12.3	79.4	44.7	.040	4.5	2.8
AIpparel	**5.4**	**85.2**	**82.7**	**.020**	**2.7**	**77.2**

AIpparel achieves state-of-the-art performance in both datasets and surpasses SewFormer-FT by a large margin on GCD-MM.

#STITCH ACC · GCD-MM

77.2% vs **2.8%**

AIpparel vs SewFormer-FT

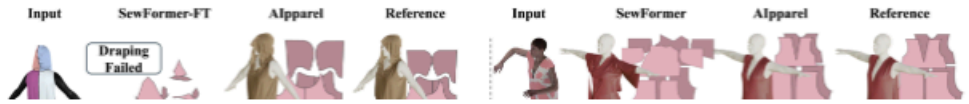


FIGURE 3 · QUALITATIVE

GCD-MM (left): AIpparel reconstructs simulation-ready patterns; SewFormer-FT fails. SewFactory (right): SewFormer produces inaccurate panels and wrong garment types; AIpparel recovers accurate patterns.

59.0% accuracy — direct multimodal reasoning, no external modules

Table 3 — AIpparel vs SewFormer-FT† vs DressCode†

TABLE 3 Multimodal Sewing Pattern Prediction

METHOD	ACCURACY ↑	PANEL L2 ↓
SewFormer-FT†	10.3	22.4
DressCode†	0.6	31.0
AIpparel	**59.0**	**6.1**

Compared to single-modal methods augmented with existing LMMs, our model outperforms both baselines by a large margin.

QUALITATIVE EXAMPLE

PROMPT

"semi-formal garden party"

DressCode† → mini-skirt

does not correlate well with the description

AIpparel → godet skirt

more appropriate for the occasion

DressCode requires external modules to translate the sewing pattern into short keywords for input, losing important information about the original garment style.

83.4% accuracy on language-instructed editing

Table 4 — AIpparel vs SewFormer* vs DressCode*

TABLE 4 · SEWING PATTERN EDITING

SOTA LMMs (GPT-4V, DALL-E) facilitate both baselines; AIpparel still outperforms by a large margin.

METHOD	ACCURACY (%)	PANEL L2 (CM)
SEWFORMER*	9.5	18.6
DRESSCODE*	37.0	13.7
AIPPAREL	83.4	1.5

+46.4 pts accuracy over the strongest adapted baseline (DressCode*).

EDITING EXAMPLES

01

“include a hood”

Tank top → hooded tank top. AIpparel adds the hood while preserving the original silhouette.

DressCode* turns the tank top into a full-length dress.

02

“make the skirt longer”

Short skirt → elongated skirt. Hemline extended, waistband and panel structure intact.

DressCode* changes the tight skirt into a flared one.

ABLATION

Tokenizer + regression objective — 25× faster, more accurate

Table 5 — DressCode vs Ours w.o. reg. vs Ours

TABLE 5 · ABLATION

Text-to-garment on DressCode's test set · averaged over 100 samples

METHOD	ACCURACY	PANEL L2	INFERENCE
DRESSCODE	38.4	22.4	52.2
OURS W.O. REG.	79.0	7.2	3.4
OURS TOKENIZER + EQ. 2	85.0	6.1	2.1

VS. DRESSCODE

25×

faster at inference

52.2 s → 2.1 s

TOKENIZER **Ours w.o. reg.** quantizes vertex positions, edge control parameters and 3D transforms into **256 bins**, predicted via next-token prediction.

TIMING Average wall-clock time to generate and decode one garment, profiled on a **single A6000 GPU**.

VISUAL ANALYSIS

Fine details preserved, instructions followed

Waistband panels, sleeve cuffs, hoods, skirt length

01 Reconstruction quality

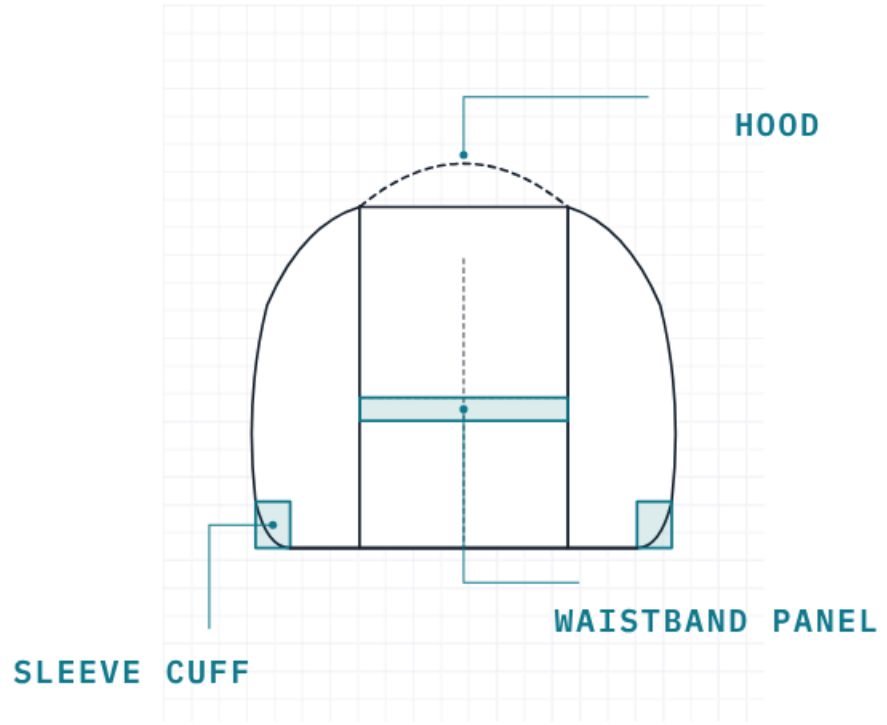
Accurately recovers sewing patterns from the image, producing superior draping simulations. Follows both visual cues (sleeve and skirt length) and textual cues (waistline, neckline).

02 Editing precision

Follows instructions precisely — adding a hood to a tank top or elongating a skirt — while preserving the original style. DressCode tends to overhaul the garment instead.

03 Tokenizer geometry

Outputs geometrically regular sewing panels aligned with the text. DressCode's tokenizer can produce unrealistic shapes that contradict the description (e.g., a "flared hem" rendered straight).



SCHEMATIC · FINE DETAILS AIPPAREL PRESERVES

LIMITATIONS

Three boundaries, three directions

Non-manifold features • Fabrication • Synthetic-to-real gap

01	**LIMITATION** The current representation is constrained to garments representable by manifold surfaces. Design elements such as pockets require non-manifold structures and remain out of scope.	→	**DIRECTION** An efficient representation that accurately models non-manifold features while staying compatible with LMMs.
02	**LIMITATION** Fabricating the generated garments is not addressed. Turning a predicted pattern into a sewable garment requires physical and material constraints we do not model.	→	**DIRECTION** Incorporate physical and material constraints directly into sewing-pattern prediction.
03	**LIMITATION** Existing sewing-pattern datasets, including GCD and its GCD-MM extension, are mostly procedurally generated. The synthetic-to-real gap to 3D-scanned garments is unbridged.	→	**DIRECTION** Curate real-world 3D-scan sewing patterns and study domain transfer across the synthetic–real divide.

BROADER IMPACTS

While we believe our model can advance AI-assisted fashion design, we acknowledge potential risks inherited from the pre-trained LLaVA model — generative AIs can spread misinformation or create biases harmful to society. We do not condone these or other improper usages of our model.

TAKEAWAYS

Three things to remember

Dataset • Tokenization • Multimodal model

01

A multimodal dataset at scale

GCD-MM extends the largest public sewing-pattern dataset with text, image, and editing-pair annotations — over a hundred thousand garments ready for multimodal pretraining.

02

A tokenization that makes LMMs efficient

Special tokens plus regression heads let us represent complex patterns in roughly 250 tokens — about a hundred times fewer than prior work, and the key to retargeting LMMs to garment prediction.

03

The first multimodal foundation model

AIpparel takes language, images, and sewing patterns as input — achieving state-of-the-art on single-modal tasks and enabling new applications such as language-driven garment editing.

"AIpparel achieves state-of-the-art performance in single-modal tasks, including text-to-garment and image-to-garment prediction, and enables novel multimodal garment generation applications such as interactive garment editing."

CONCLUSION

A scalable recipe for multimodal garment generation

AIpparel • GCD-MM • future directions

AIpparel is the first multimodal foundation model for garment sewing patterns — a 7B-parameter model fine-tuned on the GCD-MM dataset with a novel sewing-pattern tokenizer and a mixed training objective. It achieves state-of-the-art results on single-modal generation tasks and enables new applications such as language-driven sewing-pattern editing.

DATASET RELEASE **GCD-MM will be released publicly.**

01 Non-manifold features
Pockets, zippers, and other features that break the flat-panel assumption.

02 Fabrication-aware prediction
Predicting patterns that respect fabric properties and sewing constraints.

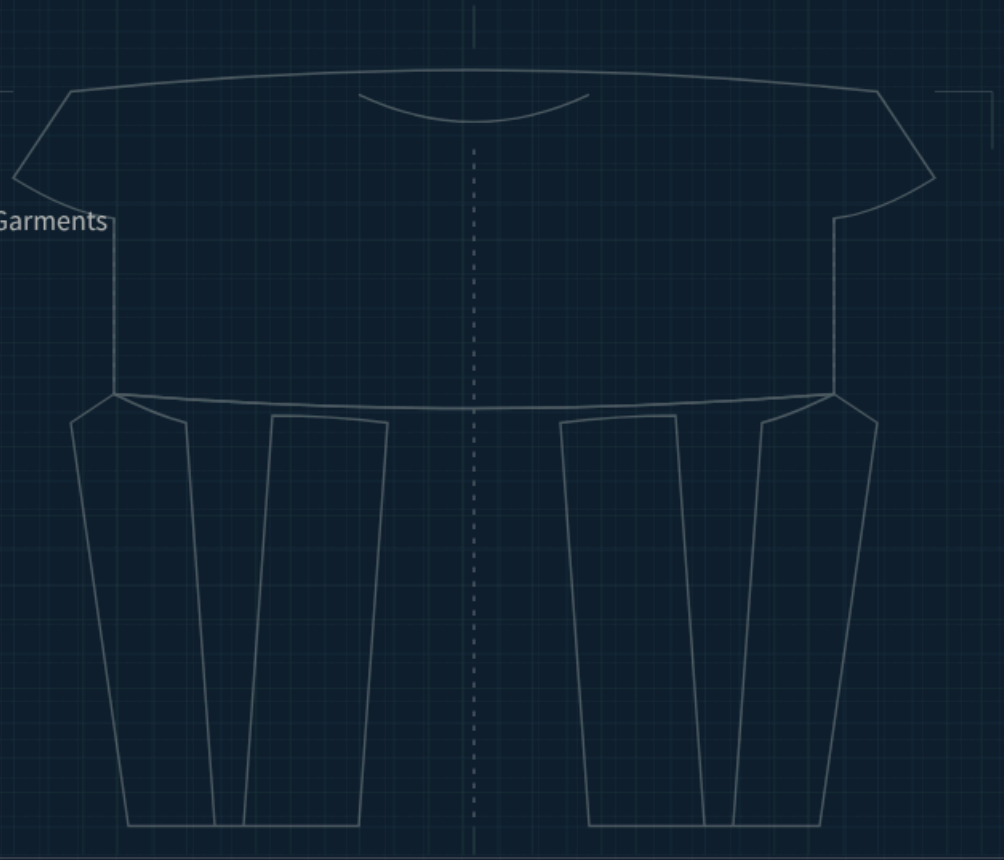
03 Synthetic-to-real bridge
Closing the gap between synthetic training data and real-world 3D scans.

> "By fine-tuning LMMs to understand sewing patterns, we take first steps towards a vision-language-garment model that transfers web knowledge to garment generation and editing."

THANK YOU

AIpparel

A Multimodal Foundation Model for Digital Garments



georgenakayama.github.io/AIpparel

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Code and the GCD-MM dataset will be released at the project website.

ACKNOWLEDGEMENTS

Supported by Google, an ERC Consolidator Grant No. 101003104 (MYCLOTH), an ARL grant W911NF-21-2-0104, a Vannevar Bush Faculty Fellowship, and LVMH.